

## **Rainfall Comparison between Seattle and Philadelphia**

### **Intro**

The purpose of this project is to analyze rainfall data for Seattle and Philadelphia to answer the following questions: Which city receives more rainfall per year, and which city has a higher proportion of rainy days per year?

### **Data**

The data analyzed in this project was sourced from the National Oceanic and Atmospheric Administration (NOAA) through the NOAA Climate Data Online (CDO) portal. The data is from two weather stations, one in Seattle and one in Philadelphia. The data spans a total of five years, from January 1, 2018, to December 31, 2022.

The data can also be accessed through the following link:

<https://www.ncei.noaa.gov/cdo-web/>

### **Analysis**

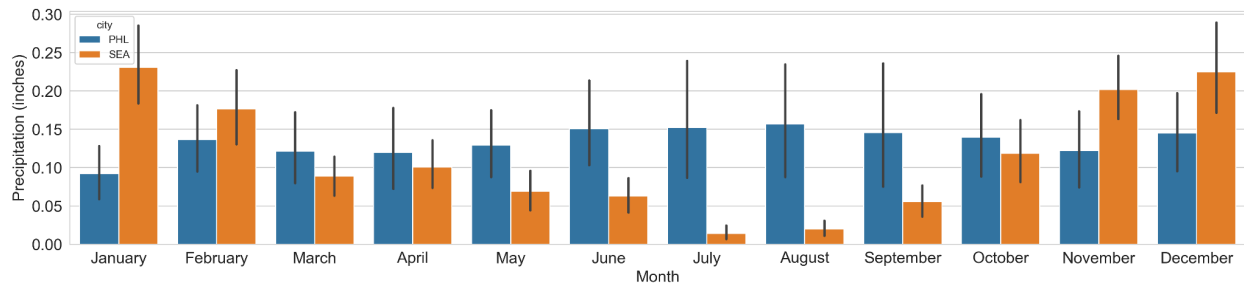
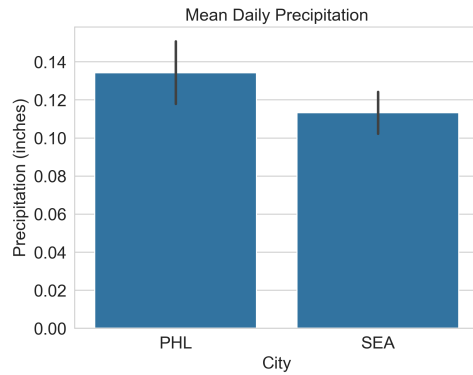
The data for this project was gathered by NOAA, however, it had to be appropriately cleaned and organized for analysis. The data that we used was the precipitation data gathered by the two weather stations, and the data was stripped of non-relevant data and reorganized into a long format. This data was then visualized and aggregated to find underlying patterns regarding the precipitation data in terms of rainfall per year and the proportion of days with rainfall. On days of the year where rainfall data was missing, the mean rainfall was added in its place.

### **Results**

#### *Comparison of Average Rainfall*

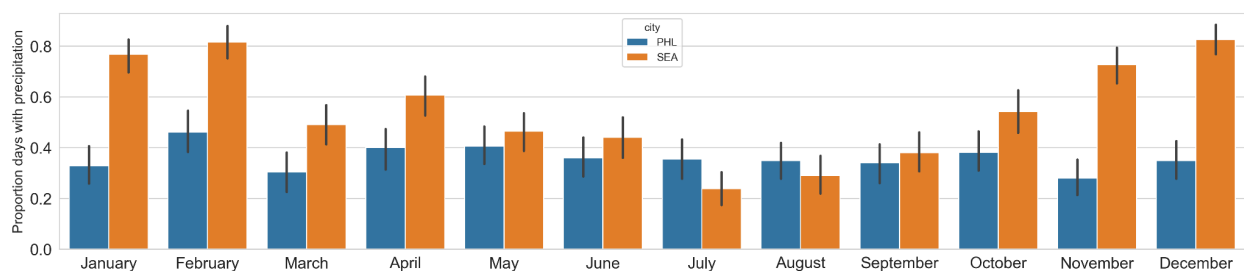
Our analysis indicated that Philadelphia received more inches of rainfall than Seattle per year, however, Seattle had a higher proportion of rainy days per year when compared to Philadelphia. Philadelphia received, on average, 0.1342 inches of rain per day, while Seattle received an average of 0.1133 inches of rain per day. This reflected that, in terms of inches of rain, Philadelphia received an average of 18.5% more rain per year throughout the 5-year span that our dataset covered. This is reflected in the visuals below.

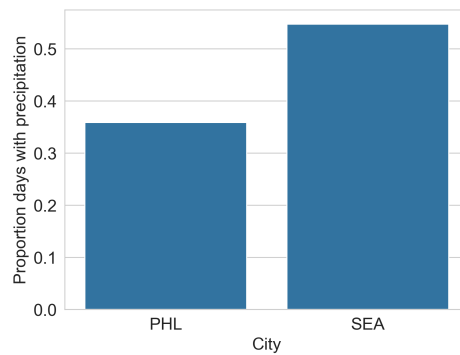
The visuals also indicate that in the month of January, Seattle averaged 0.2307 inches of rain, while Philadelphia received an average of 0.0923 inches of rain, indicating that Seattle received 150% more inches of rain per day on average in January. In the month of July, Philadelphia averaged 0.1519 inches of rain per day, and Seattle averaged 0.0140 inches of rain per day. This shows that in July, Philadelphia averaged 985% more inches of rain per day when compared to Seattle.



### *Comparison of Proportion of Rainy Days*

When viewed proportionally, on the other hand, Seattle had a higher proportion of rainy days per year when compared to Philadelphia. Throughout the entire year, 54.7% of Seattle's days had rain, while Philadelphia experienced rainy days 35.9% of the time. In January specifically, Seattle experienced rain on 76.8% of days, while Philadelphia experienced rain 32.9% of the time. On the other hand, in the month of July, Philadelphia had rainy days, on average, 35.5% of the time, while Seattle saw rain 23.9% of the time. These results are visualized below.





## Conclusion

Seattle and Philadelphia were the focus locations of this project, and the key questions that were answered were which city sees more annual rainfall in terms of inches and which city sees a higher proportion of rainy days throughout the year.

In terms of mean annual rainfall, Philadelphia averaged 18.5% more rain throughout the year. In terms of the proportion of days with rain, on the other hand, Seattle averaged 52.4% more days with rain throughout the year when compared to Philadelphia. These were yearly averages as well, and there was a lot of variety each month of the year.